REPURCHASE AGREEMENTS

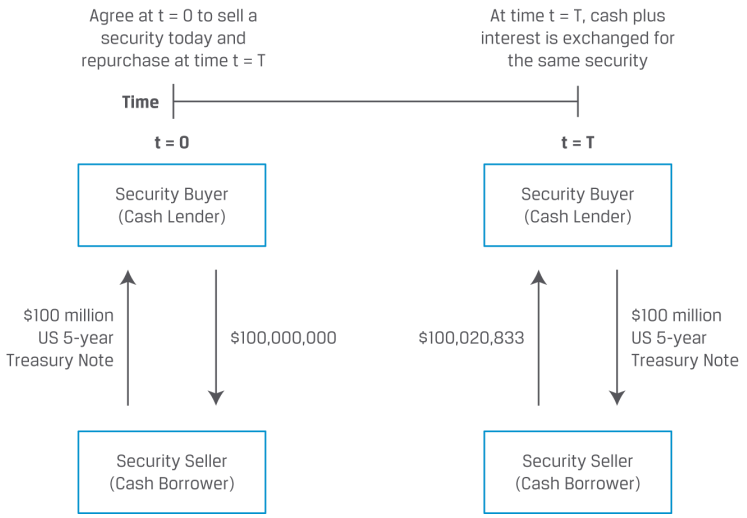
Learning Outcome

- describe repurchase agreements (repos), their uses, and their benefits and risks

An important source of *secured* short-term lending and borrowing is the repurchase agreement (repo) market. A **repurchase agreement**, or repo, involves the sale of a security with a simultaneous agreement by the seller to buy the same (or a similar) security back from the purchaser at an agreed-on **repurchase price** and future date called the **repurchase date**. Exhibit 4 shows the mechanics of a simple repurchase agreement.

Exhibit 4:

Repurchase Agreement



Assume that today (t = 0) the current US five-year Treasury note trades at a price equal to the bond's face value of USD100,000,000. The security *buyer* takes delivery of the US Treasury note today and pays the security *seller* USD100,000,000. If we assume a repo term of 30 days (and 360 days in a year) and an annual interest rate (or **repo rate**) of 0.25%, then the buyer agrees to return the Treasury note 30 days from today ($t = T$) to the seller at a repurchase price of USD100,020,833, calculated as follows:

$$\text{USD}100{,}000{,}000 \times [1 + (0.25\% \times 30/360)] = \text{USD}100{,}020{,}833.$$

In effect, the security seller borrows USD100,000,000 on a short-term basis at a low cost, with interest (USD20,833) paid at maturity, because the loan is collateralized by the US Treasury note.

The details of repo and other money market interest calculations are addressed in a later lesson. In a repo transaction, the seller of the security (cash borrower) retains ownership of the security over the repo term, as well as the interest and actual coupon paid by the security, while the security buyer (cash lender) receives the repo rate.

Because cash lenders in the repo market are primarily interested in both the liquidity and safety of their investments, most repos are typically very short-term transactions with the highest-quality securities (e.g., sovereign bonds) serving as collateral. Repo periods range from overnight to so-called **term repos**, those involving any maturity longer than one day.

a specific group of securities as eligible collateral (such as government bonds of a specific maturity) in what is known as a **general collateral repo** transaction. These transactions take place at the **general collateral repo rate**.

In addition to the high quality of underlying securities, repos include features designed to reduce the risk of a collateral shortfall over the contract life. One such feature is the provision of collateral in excess of the cash exchanged, known as **initial margin**, defined as the following ratio:

$$\text{Initial Margin} = \frac{\text{Security Price}_0}{\text{Purchase Price}_0}. \quad (1)$$

A 100% initial margin indicates a fully collateralized loan, while a higher margin indicates even greater initial collateral protection. This is alternatively considered a *reduction* or **haircut** of the underlying loan relative to the initial collateral value, which may be shown as follows:

$$\text{Haircut} = \frac{\text{Security Price}_0 - \text{Purchase Price}_0}{\text{Security Price}_0}. \quad (2)$$

The use of excess collateral is shown in the following repo example.

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Knowledge Check

1. Assume the repurchase agreement in Exhibit 4 is subject to a 102% initial margin.

Calculate the revised original purchase price (loan amount).

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Solution: Using an initial margin of 102%, we can solve for a revised purchase price using Equation 1:

$$\text{Initial Margin} = \frac{\text{Security Price}_0}{\text{Purchase Price}_0}.$$

$$102\% = \frac{\text{USD}100,000,000}{\text{Purchase Price}_0}.$$

$$\text{Purchase Price}_0 = \text{USD}98,039,216.$$

Note that security seller must finance the remaining USD1,960,784 from other sources.

2. Calculate the associated repo haircut.

Hide Solution

Solution: Using Equation 2, solve for the haircut as follows:

$$\text{Haircut} = \frac{\text{Security Price}_0 - \text{Purchase Price}_0}{\text{Security Price}_0}.$$

$$\text{Haircut} = \frac{\text{USD}100,000,000 - \text{USD}98,039,216}{\text{USD}100,000,000}.$$

$$\text{Haircut} = 1.96\%.$$

3. Calculate the repurchase price using the same 30-day repo term and 0.25% repo rate.

Hide Solution

Solution: The repurchase price may be calculated as follows:

Repurchase price = USD98,039,216× [1 + (0.25% × 30/360)].

Repos address collateral value *changes* by granting contract participants the right to request additional collateral (or release existing collateral) to maintain a security interest equal to the original initial margin terms. This variable margin payment (referred to as **variation margin**) is equal to the difference between current margin required and the security's price at time *t*, as shown in Equation 3:

$$\text{Variation margin} = (\text{Initial margin} \times \text{Purchase price}_t) - \text{Security Price}_t. \quad (3)$$

For example, a security price *decline* requires a cash borrower (security seller) to provide additional collateral to the seller.

Knowledge Check

1. Assume the same repurchase agreement terms from Exhibit 4 and the previous Knowledge Check. Calculate the variation margin five days after trade inception if the price of the five-year US Treasury note rises to 103% of the security's USD100 million face value.

Hide Solution

Solution:

Recall the original loan amount (or purchase price at *t* = 0) is equal to USD98,039,216, or USD100,000,000/1.02. We may solve for the purchase price when *t* = 5 as follows:

Purchase price = USD98,039,216× [1 + (0.25% × 5/360)].

Purchase price = USD98,042,620.

To calculate the variation margin, we use Equation 3:

$$\text{Variation margin} = (\text{Initial margin} \times \text{Purchase price}_t) - \text{Security price}_t.$$

$$\text{Variation margin} = (102\% \times \text{USD}98,042,620) - \text{USD}103,000,000.$$

$$\text{Variation margin} = -\text{USD}2,996,528.$$

The security price increase to USD103,000,000 resulted in an overcollateralization of the loan by USD2,996,528, allowing the cash borrower (security seller) to request the release of five-year US Treasury notes with a face value of approximately USD2.91 million (= USD2,996,528/1.03).

Repo contract participants typically agree to margining and other terms, such as the substitutability of collateral and events of default, in a **master repurchase agreement** or similar legal document that governs all trades between parties.

Repurchase Agreement Applications and Benefits

Financial market participants use the repo market for three specific purposes:

- Finance the ownership of a security
- Earn short-term income by lending funds on a secured basis
- Borrow a security in order to sell it short

Financial institutions that trade securities often or hold securities inventory frequently participate in the repo market as security sellers or cash borrowers. For example, say an investor sells a bond to a bank's trading desk in exchange for cash. The bank must pay the investor immediately although it may intend to sell the security to another party at a later time. To do so, the bank may borrow the purchase price from another bank or an

From a security buyer's perspective, a repo offers a short-term cash investment on a collateralized basis with minimal liquidity or default risk. Overnight cash repos with high-quality collateral result in the lowest return; investors can generate higher returns for longer repo terms or by accepting less liquid or lower quality collateral. These investors include banks and other institutions, such as mutual funds or pension funds. As shown in Exhibit 2, these short-term assets appear on a bank's balance sheet as "Securities Purchased under Agreements to Resell or Borrowed."

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Central banks are also active participants in the repo market to conduct monetary policy. While reserve requirements and the outright purchase and sale of securities are more permanent means of influencing the money supply, repo contracts are frequently used for shorter-term, transitory policy measures. For example, a central bank may temporarily increase cash reserves in the banking system by borrowing securities and lending cash.

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Sometimes, a security buyer enters a repo not only to lend cash for interest but also to use the security for another use. For instance, a hedge fund may take a short position in a security, with a view that its price will decline. The use of a repo transaction to facilitate this transaction is shown in Exhibit 5.

Exhibit 5:

Reverse Repos and Short Security Positions

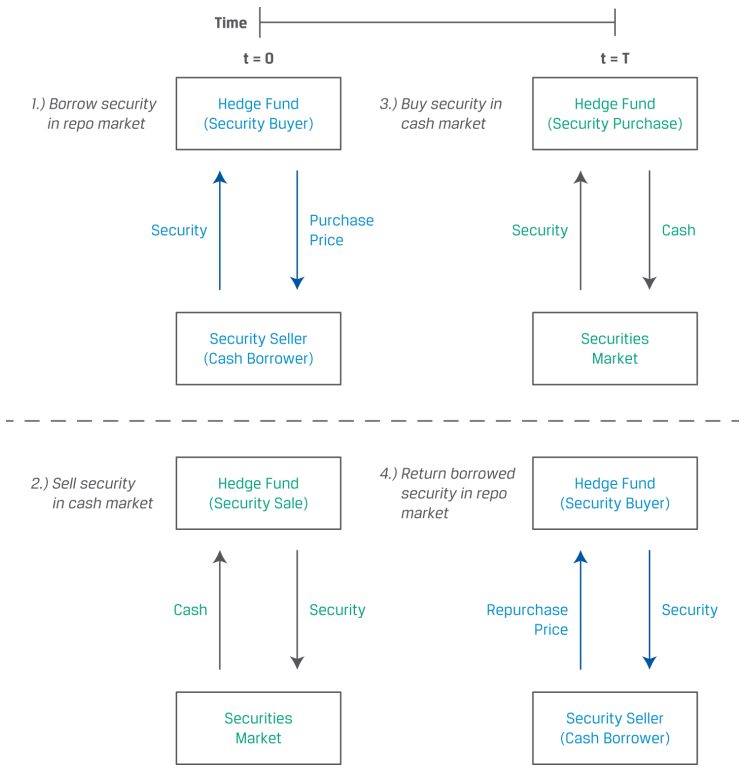

Exhibit 5 illustrates this transaction in four steps:

At time $t = 0$:

1. The hedge fund *borrows* the security in exchange for cash in a repo transaction.
2. The hedge fund simultaneously *sells* the security in the secondary market and receives cash.

   At time $t = T$:
3. The hedge fund *purchases* the security in the secondary market and pays the current market price.

is sometimes referred to as a **reverse repurchase agreement** or reverse repo.

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Knowledge Check

1. Assume that a hedge fund enters the same repurchase agreement from Exhibit 4 as the security buyer (cash lender), sells the security short in the secondary market today, and buys back the security to deliver for repo settlement in 30 days. For purposes of this example, we exclude initial or variation margin.

   Calculate the hedge fund's return on the transaction if it is able to purchase the five-year US Treasury note in 30 days in the secondary market at a price equal to 99% of the face value.

   Hide Solution

   Solution:

   Follow the four steps in Exhibit 5:

   1. The hedge fund *borrows* the five-year US Treasury note priced at USD100,000,000 today and delivers USD100,000,000 in cash to the security seller in a repo transaction.

      As in Exhibit 4, the hedge fund (security buyer) agrees to return the five-year Treasury note 30 days from today to the seller at a repurchase price of USD100,020,833, calculated (using a 0.25% repo rate) as follows:

      USD100,000,000 × [1 + (0.25% × 30)/360] = USD100,020,833.

   2. The hedge fund simultaneously *sells* the five-year US Treasury note in the secondary market today and receives USD100,000,000 in cash.
   3. In 30 days, the hedge fund purchases the five-year US Treasury note in the secondary market at the current market price of USD99,000,000.
   4. The hedge fund simultaneously *delivers* the five-year US Treasury note and receives the pre-agreed repurchase price of USD100,020,833 to settle the repo.

   The hedge fund has therefore earned a return of USD1,020,833 (= USD100,020,833 − USD99,000,000), ignoring transaction costs. The hedge fund's short position yields a profit because the price of the Treasury note fell over the time period.

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Ignoring initial margin and other transaction costs, the hedge fund will earn a positive return if the short position gain exceeds the repo rate for the period. Note that such a repo transaction requires delivery of a particular security, known as a special trade, which occurs at a special collateral rate. When a particular security is subject to very high borrowing demand, this special repo rate may fall below the general collateral repo rate or even below zero. Under a negative repo rate, the security buyer *pays* interest on cash lent to the cash borrower.

Repo rates are influenced by several factors, including the following:

- Money market interest rates: Repo rates are correlated with other short-term interest rates. Central banks use secured repo markets to influence the unsecured central bank funds rate.
- Collateral quality: Higher collateral risk increases repo rates. Repo rates are usually higher for equity securities or emerging market bonds than for developed market government bonds.

- Collateral uniqueness: The higher the demand for a specific security, the lower the repo rate. The most recently issued or on-the-run developed market sovereign bonds typically have the lowest repo rates in a given market.
- Collateral delivery: Repo rates are usually higher when either the cash lent is undercollateralized or no collateral is provided to the funds lender.

Risks Associated with Repurchase Agreements

Repo markets are a widely used source of short-term funding, with relatively low borrowing costs relative to other financing available to banks and other financial institutions. However, repurchase agreements involve a number of structural risks, and excessive reliance on this form of secured financing under adverse market conditions can lead to financial distress or insolvency.

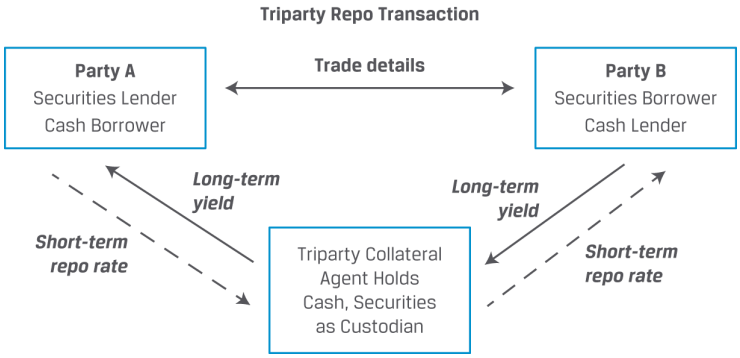
Each repo contract participant is exposed to the risk that the other party is unable to meet its obligations. While the secured nature of repo contracts reduces risk due to the value of the underlying collateral, several risks are important to consider:

- Default risk: Default risk is the primary exposure under a repo transaction, despite the existence of collateral. Collateral received from less creditworthy parties is more likely to be tested under a default scenario and may face illiquidity, adverse price changes, and legal or operational challenges.
- Collateral risk: In addition to minimizing liquidity and credit risks to ensure quick and efficient liquidation under a default, the collateral should have little or no correlation with the credit risk of the repo counterparty in order to diversify credit exposure.
- Margining risk: Proper and timely collateral valuation and transfer of variation margin to minimize collateral shortfalls in the event of liquidation following a default are important. Such adverse market conditions may also cause large changes in collateral value, increasing margin calls among market participants and forcing further liquidations.
- Legal risk: Addresses the ability to enforce legal rights under a repurchase agreement.
- Netting and settlement risk: Addresses the ability of repo contract participants to both offset or net the obligations of a non-defaulting party and take possession of collateral or cash in settlement of a trade.

In many cases, repo market participants manage these risks by engaging a third party. Repos executed directly between two parties are called bilateral repos; under a **triparty repo**, both parties agree to use a third-party agent for the transaction. While triparty agents do not change the credit risk relationship between repo participants in the event of default, they create cost efficiencies by providing access to a larger collateral pool and multiple counterparties, as well as specializing in the valuation and safekeeping of assets. The triparty agent (a custodian or clearinghouse) administers the transaction and is responsible for cash, securities, collateral valuation, and management, as well as collateral custody, as shown in Exhibit 6.

Exhibit 6:

Bilateral versus Triparty Repo Agreements



While the repo market represents a more stable funding source than short-term unsecured financing, both the uncommitted nature of repo markets and their very short (mostly overnight) maturity give rise to significant rollover and liquidity risks under adverse market conditions. Financial institutions must balance the low cost of repo funding with the greater financial flexibility of more costly longer-term financing alternatives, such as long-term debt and equity. In theory, the collateralized nature of repo transactions should provide cash lenders comfort to roll existing repo trades. In practice, financial firms that have lost the confidence of market participants have faced substantial losses and even bankruptcy during past financial crises due in part to their heavy reliance on repo financing.

The Fall of Bear Stearns and the Repo Market

Bear Stearns was the fifth-largest US investment bank, with USD350 billion in total assets at the end of 2006 and record revenue of over USD9 billion in that year. As a major underwriter of mortgage-backed securities (MBS) and subprime MBS, the firm faced challenges in 2007 as the MBS market deteriorated. The firm's losses mounted due to the failure of subprime mortgage hedge funds it had sponsored in July 2007 and the write-down of over USD2 billion in MBS it held later in the year, resulting in its first ever quarterly loss and a credit downgrade at the end of 2007.

At the same time, Bear Stearns significantly reduced its unsecured short-term funding (from USD25.8 billion at the end of 2006 to USD11.6 billion at the end of 2007) and specifically its commercial paper (USD20.7 billion to USD3.9 billion) in favor of secured funding–specifically, repo borrowing. As of 10 March 2008, Bear Stearns had USD18 billion in cash, but within two days, a loss of confidence due to the credit downgrade of related entities and market rumors caused this to fall to USD2 billion. Specifically, repo counterparties declined to roll over or renew repo trades, prime brokerage clients withdrew cash and securities, and other banks withdrew or refused to extend credit. On Friday, 14 March 2008, Bear Stearns received an emergency Federal Reserve loan against its securities collateral to prevent its failure, and on Monday, 17 March 2008, Bear Stearns agreed to be purchased by JPMorgan, which received a loan from the Federal Reserve.

Question Set

1. Assume that today (t = 0) the current US five-year Treasury note trades at a price equal to the bond's face value of USD100,000,000. The security buyer takes

B. 2.91%.
C. 3.00%.

Hide Solution

Solution:

B is correct. Using an initial margin of 103%, the purchase price (loan amount) is calculated as USD100,000,000/1.03 = USD97,087,379.

The repo haircut is then calculated as

(USD100,000,000 − USD97,087,379)/USD100,000,000 = 2.91%.

2. Assume that today (t = 0) the current US five-year Treasury note trades at a price equal to the bond's face value of USD200,000,000. The security buyer takes delivery of the US Treasury note today and pays the security seller a purchase price based on an initial margin of 102%. Assume a repo term of 15 days (and 360 days in a year) and an annual interest rate (or repo rate) of 0.50%. If the buyer agrees to return the five-year Treasury note 15 days from today (t = T) to the seller, the repurchase price is closest to:

A. USD196,119,281.
B. USD196,160,130.
C. USD200,041,667.

Hide Solution

Solution:

A is correct. Using an initial margin of 102%, the purchase price (loan amount) is calculated as USD200,000,000/1.02 = USD196,078,431.

The repurchase price is then calculated as follows:

USD196,119,281 = USD196,078,431 × [1 + (0.50% × 15)/360].

B is incorrect because it incorrectly uses a repo term of 30 days:

USD196,160,130 = USD196,078,431 × [1 + (0.50% × 30)/360].

C is incorrect because it does not incorporate the initial margin in calculating the purchase price:

USD200,041,667 = USD200,000,000 × [1 + (0.50% × 15)/360].

3. Which of the following statements about repos is correct?

A. Financial market participants use the repo market to earn short-term income by lending funds on a secured basis.
B. Investors can generate higher returns for shorter repo terms.
C. From a security buyer's perspective, a repo offers a short-term cash investment on a collateralized basis with significant liquidity or default risk.

Hide Solution

Solution:

A is correct. Financial market participants use the repo market to earn short-term income by lending funds on a secured basis. B is incorrect because investors can generate higher returns for longer (not shorter) repo terms. C is incorrect because from a security buyer's perspective, a repo offers a short-term cash investment on a collateralized basis with minimal (not significant) liquidity or default risk.

4. Which of the following statements about repos is correct?

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have the highest repo rates in a given market.

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Solution:

A is correct. Repo rates are correlated with other short-term interest rates. Central banks use secured repo markets to influence the unsecured central bank funds rate. B is incorrect because repo rates are usually higher for equity securities or emerging market bonds than for developed market government bonds. C is incorrect because the most recently issued or on-the-run developed market sovereign bonds typically have the lowest repo rates in a given market.

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